INTERNET CONCEPT AND WEB DESIGN(C5001)

UNIT 1: INTERNET BASICS

Basic concepts, Communication on the Internet, Internet Domains, Internet Server Identities, Establishing Connectivity on the Internet, Client IP Address, A Brief Overview of TCP/IP and its Services, Transmission Control Protocol, Web Server , Web Client, Domain Registration. **[Q-2]**

UNIT 2: JAVA SCRIPT

Java Script in Web Pages, Advantages of Java Script, Advantages of Java Script, Data Types and Literals, Type Casting , Java Script Array, Operators and Expression, Conditional Checking , Function, User Defined Function. **[Q-2]**

UNIT 3: UNDERSTANDING XML

SGML, XML, XML and HTML, Modeling XML Data, Styling XML with XSL, XHTML. **[Q-2]**

UNIT 4: CREATION OF DYNAMIC WEB PAGES USING JSP

Dynamic Web Page, Introduction of JSP, Pages Overview, JSP Scripting, Standard Action, Page Directive, Include Directive. **[Q-1]**

UNIT 5: PHP

PHP installation and Introduction, Loops, String Functions in PHP, PHP Email Function, PHP Basics, Variables, Arrays in PHP with Attributes, Date & Time, Image Uploading. **[Q-1]**

INTERNET CONCEPT AND WEB DESIGN

DESIGN AND ANALYSIS OF ALGORITHMS(C5002)

UNIT 1: INTRODUCTION & DESIGN OF EFFICIENT ALGORITHM
Algorithm, Complexity, Asymptotic Notations, Solving recurrences. **[Q-1]**

UNIT 2: DIVIDE AND CONQUER
Binary Search, Finding Maximum and Minimum, Merge Sort, Quick Sort. **[Q-2]**

UNIT 3: THE GREEDY METHOD:
The General Method, Minimum Cost Spanning Trees: Kruskal & Prism's Algorithm. **[Q-2]**

UNIT 4: DATA STRUCTURE FOR SET MANIPULATION PROBLEMS
Fundamental Operations on Set, Hashing Technique, Binary Search Trees. **[Q-2]**

UNIT 5: ALGORITHM ON GRAPHS
Depth First Search, BFS, Depth First Search of a Directed Graph. **[Q-1]**

DESIGN AND ANALYSIS OF ALGORITHMS

LINUX PROGRAMMING (C5003)

UNIT 1: Overview of Linux Architecture:
Kernel, Processes, Time Sharing, Shell, Files and Directories, Creation of A File, Inode Numbers and Filenames, File Security, File Systems, Peripheral Devices as Files. **[Q-2]**

UNIT 2: Linux Editor and Basic LINUX Commands:
Ed Editor, Vi Editor, Redirections, Piping, Filters, LINUX Utilities, Grep, Sed, Awk etc. **[Q-1]**

UNIT 3: Introduction to Shell Scripts:-
Bourne Shell C Shell, Shell Variables, Scripts, Metacharacters and Environments, If and case Statements. For, While and until Loops. **[Q-2]**

UNIT 4: Awk Programming:-
Awk Pattern, Scanning And processing Language, Begin and End Patterns. Awk Arithmetic and variables. Awk Built in Variable names and Operators. Arrays and Strings. **[Q-1]**
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UNIT 5: Introduction To System Administration:-
File System ,System Administrator and its Role, Function of a system Manager, practical Aspect of System Administrator, Visual Attributes.System Call and C function Library:- Linux System Call library function and Math Library, standard I/O package. File Hindling, Command Lie parameters, Linux –C interface, C files.
[Q-2]

LINUX PROGRAMMING

COMPUTER ORIENTED NUMERICAL METHOD (C5004)

UNIT 1: ERRORS IN NUMERICAL CALCULATIONS

Numbers and their accuracy, Errors and their Computations- Absolute, Relative and Percentage, General Error Formula. **[Q-2]**

UNIT 2: SOLUTION OF ALGEBRAIC AND TRANSCENDENTAL EQUATIONS

Introduction, Bisection method, Iteration method, Method of False Position, Newton- Raphson method. **[Q-1]**

UNIT 3: INTERPOLATION

Introduction, Errors in Polynomial Interpolation, Finite Differences-Forward, Backward and Central, Detection of errors using Difference tables, Differences of a Polynomial, Newton's formulae for Interpolation, Central Difference Interpolation Formulae- Gauss's Central Difference Formula, Interpolation with unevenly spaced points, Lagrange's Interpolation Formula, Divided Differences and their properties- Newton's General Interpolation Formula. **[Q-2]**

UNIT 4: NUMERICAL DIFFERENTIATION AND INTEGRATION

Introduction, Numerical Differentiation and Errors, Numerical Integration – Trapezoidal Rule, Simpson's 1/3 Rule, Simpson's 3/8 Rule. **[Q-2]**

UNIT 5: NUMERICAL SOLUTION OF LINEAR SYSTEM OF EQUATIONS

Direct Methods- Matrix Inversion Method, Gauss-Jordan Method, Gauss Elimination Method, Iterative Method- Gauss- Jacobi Method, Gauss-Seidel Method. Taylor's Series, Euler's method, Modified Euler's method, Runge-Kutta method of 2nd and 4^{th} order. **[Q-1]**

COMPUTER ORIENTED NUMERICAL METHOD

E-COMMERCE (E5007)

UNIT 1: INTRODUCTION TO E-COMMERCE

E-commerce, E-commerce business models, Major business-to-consumer (B2C) business models, Major business-to-business (B2B) business models, Business models in emerging Ecommerce areas, How the internet and the Web change business. **[Q-2]**

UNIT 2: E-COMMERCE INFRASTRUCTURE

The Internet, Technology background, The world wide web. A systematic approach, choosing server software, choosing the hardware for an E-commerce site, other E-commerce site tools. **[Q-2]**

UNIT 3: SECURITY AND ENCRYPTION

The E-commerce security environment, Security threats in the E-commerce environment, Technology solutions, Policies, Procedures and Laws. **[Q-2]**

UNIT 4: E-COMMERCE PAYMENT SYSTEMS

Payment systems, Credit card E-commerce transactions, E-commerce digital payment systems in the B2C arena, B2B payment systems. **[Q-1]**

UNIT 5: ETHICAL, SOCIAL, AND POLITICAL ISSUES IN E-COMMERCE

Understanding ethical, social, and political issues in E-commerce, Privacy and information rights, Intellectual property rights, Governance, Public safety and welfare. **[Q-1]**

E-COMMERCE